Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/27/2006
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Project Title:	Evolution of Local Microstructure: Spatial Temporal Correlation in Clusters Undergoing 2-Dimensional Diffusion		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3590	**Congressional District:**	21
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	02/01/2004	**End Date:**	09/30/2006
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Frazier, D. (NASA Marshall Space Flight Center) Rogers, J. (NASA Marshall Space Flight Center) Witherow, W. (NASA Marshall Space Flight Center)		
Grant/Contract No.:	NNM04AA26G, NAG8-1468		
Performance Goal No.:			
Performance Goal Text:			
	Diffusion-limited capillarity-driven coarsening of precipitates is an important and intensively studied phenomenon. Phase coarsening can occur among microstructural constituents ranging from the primary phases to widely dispersed precipitates. The wide range of volume fractions, Vv, encountered in phase coarsening makes it essential to have a fundamental theory, or model, of phase coarsening that treats the volume fraction as an input parameter of the microstructure. The classic coarsening theory developed by Todes and Lifshitz and Syozov, (TLS theory, often also termed LSW theory) is limited to infinitesimally small volume fractions, so Vv = 0. This highly idealized model proved useful to explain experimentally observed scaling time dependency of the characteristic size of precipitates, and approximates their distribution in radii. However, TLS theory neglects all particle interactions and, as a consequence, neither describes the volume fraction dependence of the coarsening rate, nor any spatial instabilities that may develop in		

Task Description:

the size distribution. This task extends modeling and holographic imaging to compare coarsening rates in low-volume-fraction microstructures by observing mixed-dimensional droplets (spherical caps on a surface coarsening by two-dimensional diffusion) during ground-based investigations. The associated microgravity objective is to study three-dimensional coarsening in a finite, but large cluster of droplets containing different volume fractions located at the central volume of the observation cell. Our experimental objectives are to study three-dimensional Ostwald ripening among finite clusters of precipitates in microstructures comprised of a dispersed liquid-liquid systems. The ELMS concept employs a new approach of studying spatial instability and self-structuring within an initially homogeneous arrangement of a finite number of spherical precipitates. Modeling and optical holography are the two key tools to be integrated for making progress in this area of materials science. Laboratory experiments on three-dimension particles supported on a two-dimensional substrate have been carried out under terrestrial conditions. Long-duration experiments in three-dimensions will require microgravity available for long periods on ISS to insure stability of the microstructure, thereby allowing more stable structures to be recorded over longer periods of time. The ELMS task, if accomplished, should enhance our understanding of microstructural systems coarsening by diffusion. Specifically, this task addresses the effects of finite clusters of droplets, as opposed to coarsening in “infinite continua.” Our tasks are to record by holography the “history” of each individual droplet within a cluster and determine the time-evolution of the entire system for periods of up to several months. We compare experiments and numerical predictions for individual droplets and for the system as a whole. Finally, we plan to study, numerically and experimentally, local effects resulting from particle-particle interactions and the possibility of spatial instabilities leading to macro-inhomogeneities in the two-phase structure.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

A new multiparticle interaction model has been Fortran coded, tested, and applied in several systems. Larger scale tests of the microstructural model are currently underway. It appears that standard series codes for inverting mathematical matrices become inefficient and ultimately fail as the number of matrix elements exceeds about 107 in the case of system having 4000 particles. Microstructures containing many particles may be evolved by stepping in time, applying the computed growth rates to each step. To overcome this difficulty, parallel coding of multiparticle diffusion is being developed and will be tested soon. A workstation cluster located at Rensselaer will support this testing.
Rensselaer’s ELMS team concentrated on both the theoretical and computational aspects of ELMS. Theoretical study of interactions and fluctuation during microstructural evolution yielded important research results. We approximated phase coarsening dynamics in an alloy system as a mesoscopic distribution of “active” diffusion sources and sinks. In the framework of a Poisson diffusion field, we showed that the interaction length scale is related to the microstructure’s volume fraction and particle size distribution. We suggested that the interaction length of a two-phase microstructure could be measured by direct experiment. This, we believe, is the first time that the interactions occurring during phase coarsening–albeit in a dilute system–have been described using fundamental physical interactions. We developed diffusional screening theory for phase coarsening, which is published in Physical Review E. The theory has been tested recently using computer simulation and comparing with experimental results. It was found that our theoretical and simulation results are in good agreement with Mahalingam et al.’s (1987) experimental results on the binary Al-Li alloy system. Our results will be compared with other experimental results.

Dr. Ke-Gang Wang designed, programmed, and implemented a multiparticle diffusion code to simulate microstructural evolution. An important finding obtained from these simulations is that there exists “noise” in the local kinetic rate associated with each microstructural environment, or “locale” during evolution. Mean-field theory treats the growth rates of particles as a deterministic kinetic process that depends only on the size of a particle relative to the average size of the entire microstructural population. From our recent simulations, however, we found that the volume fluxes of particles are different, even for particles with the identical radii, when located in different local environments. We observed that the fluctuation band of the noise for a particle’s volume flux broadens with the increase of a microstructure’s volume fraction. In addition, we found that larger particles experience stronger fluctuations, or locale noise, in their growth rates than do smaller particles. The non-linear character of a particle’s volume flux versus particle size increases steadily with increasing volume fraction. In order to describe the effect of the noise on the growth rate of particles, we proposed that multiplicative noise to be added into the general growth rate function. It was found that the multiplicative nose provides a useful approximation to describe the findings of simulation. These results were published by the Rensselaer investigators in several papers and presentations.

Our studies show that stochastic effects are always present in real biphase microstructures, and play important roles in their microstructure evolution. The stochastic growth rates of particles is definitely established, and an associated Fokker-Planck equation rather than the usual continuity equation for particle size distribution is also established. We also used mathematical techniques based on approximations to solve the Fokker-Plack equation and got PSD’s for specific volume fractions. It was found that the microstructural “noise” broadens the PSD and reduce the peak value of the PSD. These results are published in Physical Review E and presentations. This, we believe, is the first time that a systematic study was carried out on stochastic effects occurring during phase coarsening. Therefore we advanced the understanding of phase coarsening from LSW theory to our diffusion screening theory, and to stochastic theory. The new research provides new fundamentally based approaches to study the discrepancy between theory and experiment in phase coarsening.

We also extended the use of diffusional screening theory and multiparticle diffusion simulation to study polymer materials. Specifically, the authors proposed a computational strategy that permits the modeling of simultaneous interactions among a very large numbers of polymer aggregates. This modeling approach can serve as the basis of conducting “virtual” combinatorial experiments and high throughput screening of pore size distribution for ultrafiltration membranes. This result was published recently in Macromolecular Rapid Communications.

Precipitation hardening in an Fe-12Ni-6Mn maraging steel during overaging is quantified. First, applying our recent diffusion screening theory of phase coarsening, and incorporating the Ashby--Orowan relationship, we link quantifiable aspects of the microstructures of these steels to their mechanical properties, including especially the hardness. Specifically, hardness measurements allow calculation of the precipitate size as a function of time and temperature through the Ashby—Orowan relationship. Second, calculated precipitate sizes and thermodynamic data determined with Thermo-Calc are used with our recent diffusion screening phase coarsening model to extract diffusion coefficients during overaging from hardness measurements. Finally, employing more accurate diffusion parameters, we determined the hardness of these alloys independently from theory, and found agreement with experimental hardness data. The

methodology can be applied to other systems for establishment between microstructure and properties.

Employing multiparticle diffusion methods simulated three-dimensional microstructural evolution at various volume fractions. Local environmental information and particle interactions within microstructures are captured in these simulations. Consequently, this study reveals spatial, temporal, and nearest-neighbor correlations. Our study also characterizes both the distribution of interparticle spacings in polydispersed systems along with the length scales that develop during microstructural evolution. The progression of some aspects of three-dimensional microstructural evolution is presented as visualization.

During the reporting period, the ELMS team at MSFC further defined how we would like to prepare substrates for producing homogeneous arrays of droplets for the initial microstructure. Several ideas have been pursued at Huntsville, but without any new information as to whether or not ELMS would be allowed to proceed as a flight program, we have settled for designing possible alternative useful approaches without extensive and costly follow-on laboratory work. The current method for initialization of the fluid-fluid microstructures include chemically etching pristine quartz surfaces, using lithographic techniques, with homogeneously arranged surface defects as heterogeneous nucleation sites. Preparation of homogeneous array patterns on quartz are currently underway at MSFC. Experiments will use these patterned surfaces as the nucleation surfaces in cells subsequently to be fabricated to contain succinonitrile-water solutions for quenching into the miscibility-gap to form the biphase microstructures.

OVERVIEW for FINAL REPORT

This research, summarized herein in this final technical report, focused on developing theoretical and computer simulational study of local microstructural evolution. Theoretically, using statistical mechanics method, we established the relationship of maximum size in the system to the volume fraction V(V). Those localized events that may affect material properties in ways not predictable by current "mean-field" knowledge, were studied by computer simulation. From our simulation, we carefully studied the spatial correlation in phase coarsening, and clarified the expression for interparticle spacings in disperse particle system.

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